Introduction to Information Retrieval

Lecture 1: The IR Process, Vector Space Model & Boolean Retrieval

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Information Retrieval

February 4, 2026

Course Overview

1. **Introduction to IR: Vector Space Model Boolean Retrieval** $\leftarrow$ Today
2. Indexing: Sparse and Dense
3. Ranking: BM25, Query Expansion, Language Models
4. Embeddings Transformer-based Re-rankers
5. Dense Retrieval
6. Learning to Rank

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Today's Goals:

- Understand what IR is and why it matters
- Learn the fundamental IR process
- Master Boolean retrieval and the vector space model
- Build your first search engine

Outline

What is Information Retrieval?

Definition

Information Retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).

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Key Characteristics

- **Unstructured data:** Text, images, audio (vs. databases with structured records)
- **Large scale:** Billions of documents (web search, enterprise search)
- **Information need:** User wants to learn something, not exact matching
- **Relevance:** Not all results equally useful

IR vs. Database Systems

Aspect	Database Systems	IR Systems
Data	Structured (tables, schemas)	Unstructured (text, documents)
Queries	Precise (SQL)	Vague (natural language)
Matching	Exact	Approximate/Ranked
Result	All matching records	Top-k relevant documents
Failure	No results or all results	Ranked list (best effort)
Example	`SELECT * WHERE age>30`	"climate change effects"

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Key Difference

Databases: *"Give me all records matching X"*
IR: *"Give me the most relevant documents about X"*

Real-World IR Applications

Web Search

- Google, Bing, DuckDuckGo
- Billions of web pages
- Sub-second response time

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- eBay, Etsy
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Specialized Domains

- Legal document search (LexisNexis)
- Academic papers (Google Scholar)
- Medical records (PubMed)
- News archives

IR is everywhere you search for information!

The Scale of Modern IR

Web Search Statistics (2024)

- **Google:** Processes 8.5+ billion searches per day
- **Index size:** 100+ trillion web pages indexed
- **Freshness:** New pages indexed within hours
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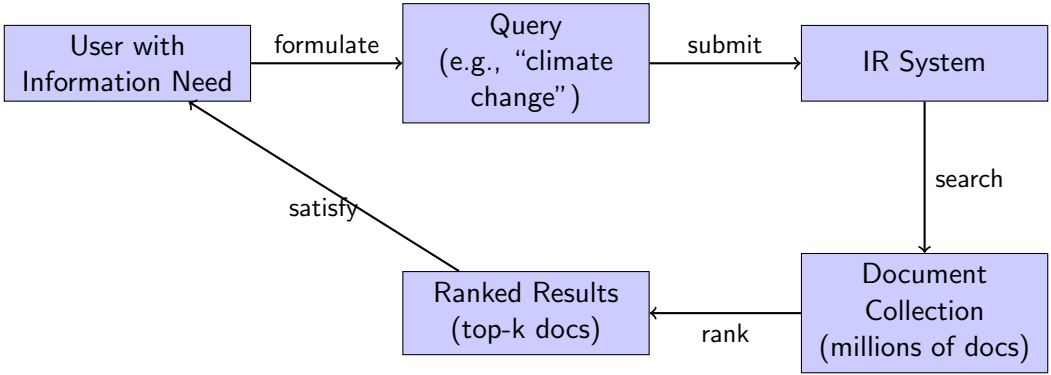
Challenges at Scale

- **Storage:** How to store 100 trillion documents?
- **Speed:** How to search in milliseconds?
- **Relevance:** How to rank billions of results?
- **Freshness:** How to keep index up-to-date?
- **Quality:** How to fight spam and misinformation?

The Information Retrieval Problem



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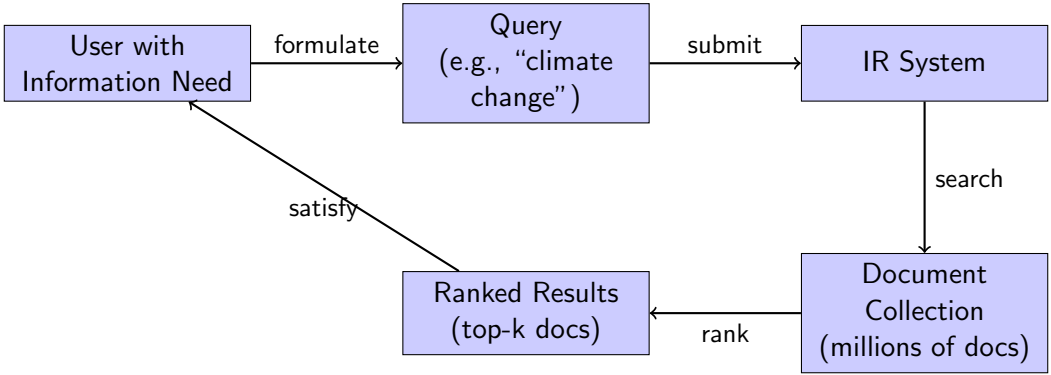
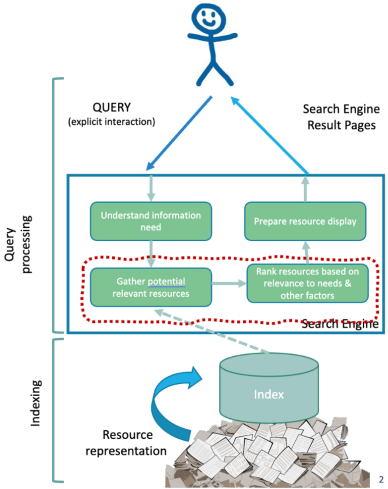


Core Questions:

1. How to represent documents and queries?
2. How to match queries to documents efficiently?

The Complete IR Pipeline



Step 1: Text Preprocessing

Goal: Convert raw text into a normalized, searchable form

Tokenization

Split text into individual tokens (words)

- Input: "The quick brown fox jumps over the lazy dog."
- Output: ["The", "quick", "brown", "fox", "jumps", "over", "the", "lazy", "dog"]

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Challenges

- Punctuation: "don't" → "don" + "t" or "dont"?
- Hyphens: "state-of-the-art" → one token or multiple?
- Numbers: "$1,000.50" → keep as is or normalize?
- URLs, emails: "http://example.com" → one token?
- Different languages: Chinese/Japanese no spaces between words

Step 1: Text Preprocessing (continued)

Normalization

Convert tokens to a standard form

- **Lowercasing:** "Apple" $\rightarrow$ "apple"
- **Accent removal:** "café" $\rightarrow$ "cafe"
- **Date/number normalization:** "5/10/2024" $\rightarrow$ "2024-10-05"

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Stop Word Removal

Remove common words with little meaning

- Stop words: "the", "a", "an", "is", "of", "in", ...
- Before: "the quick brown fox"
- After: "quick brown fox"
- **Caution:** Can hurt phrase queries ("to be or not to be")

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Many systems now *keep* stop words (storage is cheap, they can be important for some queries)

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Goal: Reduce words to their base form

Stemming

Crude chopping of word endings

- **Algorithm:** Porter Stemmer
- running $\rightarrow$ run
- runs $\rightarrow$ run
- runner $\rightarrow$ runner
- university $\rightarrow$ univers

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Lemmatization

Linguistic reduction to dictionary form

- **Uses:** Dictionary lookup
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- **Lemmatization:** Better for precise matching, NLP tasks
- Modern neural methods often skip both (learn representations directly)

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Preprocessing Pipeline Example

Original Text

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The Quick Brown Fox Jumps Over The Lazy Dog!
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Tokenization:
[The, Quick, Brown, Fox, Jumps, Over, The, Lazy, Dog, !]

Lowercasing:
[the, quick, brown, fox, jumps, over, the, lazy, dog, !]

Punctuation Removal:
[the, quick, brown, fox, jumps, over, the, lazy, dog]

Stop Word Removal:
[quick, brown, fox, jumps, lazy, dog]

Stemming:
[quick, brown, fox, jump, lazi, dog]

Step 2: Building the Inverted Index

Core Data Structure: Maps each term to documents containing it

Simple Example

Documents:

- Doc1: "the cat sat on the mat"
- Doc2: "the dog sat on the log"
- Doc3: "cats and dogs are friends"

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Term	Posting List (doc IDs)
cat	[1]
sat	[1, 2]
mat	[1]
dog	[2, 3]
log	[2]
cats	[3]
friends	[3]

Query "dog": Look up posting list → return [2, 3]

Inverted Index with Term Frequencies

Enhancement: Store not just document IDs, but also term frequencies

Term	Posting List [(docID, freq)]
cat	[(1, 1)]
sat	[(1, 1), (2, 1)]
mat	[(1, 1)]
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the	[(1, 2), (2, 2)]

Why store frequency?

- Enables ranking (documents with term appearing more are often more relevant)
- Foundation for TF-IDF, BM25 (Lecture 3)

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Step 3: Query Processing

Apply same preprocessing to queries as documents

Example

User Query: "Running Dogs in the Park"

1. **Tokenization:** [Running, Dogs, in, the, Park]
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Same preprocessing for documents and queries!

- Otherwise terms won't match
- If docs are stemmed, queries must be stemmed

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Example Queries

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- `cat OR dog` → docs containing "cat" or "dog" or both
- `java NOT coffee` → docs with "java" but not "coffee"
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Boolean Retrieval with Sets

Documents as Sets of Terms

Document Collection:

- D1: {cat, sat, mat}
- D2: {dog, sat, log}
- D3: {cat, dog, run}
- D4: {bird, fly, sky}

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Inverted Index:

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Boolean Query Processing Algorithm

```
Input: Boolean query (e.g., "A AND B OR C")
Output: Set of matching document IDs

Parse query into expression tree
for each term in query do
    Fetch posting list from inverted index
end for
Evaluate expression bottom-up:
    - For AND: Intersect posting lists
    - For OR: Union posting lists
    - For NOT: Remove from universe
return Final set of document IDs
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Optimization: Process smaller posting lists first

- `rare AND common`: Process "rare" first (smaller list)
- Intersection size $\leq$ min(—list1—, —list2—)

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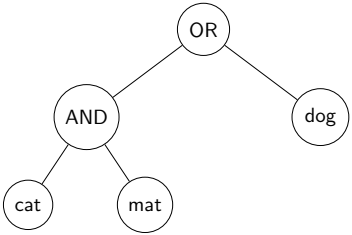

Execution:

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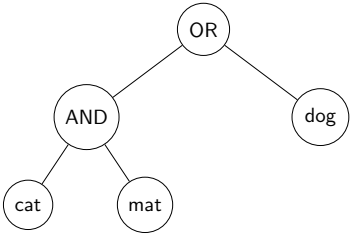

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Advantages of Boolean Retrieval

✓ Advantages

- ✓ **Precise control:** Users specify exactly what they want
- ✓ **Transparent:** Clear why documents matched
- ✓ **Efficient:** Fast set operations on posting lists
- ✓ **Predictable:** Same query always returns same results
- ✓ **Expressive:** Can construct complex queries

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- Legal document search (very precise requirements)
- Patent search (need exact specifications)
- Database-like queries on text
- Expert users who know what they want

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✗ **All or nothing**

- `climate AND change AND effects`: No results if one term missing
- Too restrictive or too permissive

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This motivates the Vector Space Model!

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Key Insight

Instead of documents being sets of terms (in/out), represent them as **vectors** with weights for each term

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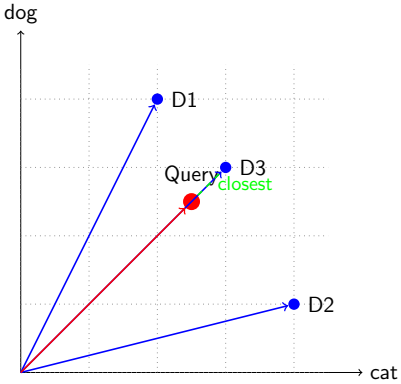
Vector Space Model: Geometric View

Documents and queries as vectors in term space



Similarity = Angle between vectors
(Smaller angle = more similar)

Term Weighting in Vector Space Model

Question: How to assign weights to terms?

Option 1: Binary Weights

$$w_{t,d} = \begin{cases} 1 & \text{if term } t \text{ appears in doc } d \\ 0 & \text{otherwise} \end{cases}$$

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Idea: Balance term frequency with term rarity

TF-IDF Formula

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where:

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Collection: 10,000 documents

Term	tf(t,d)	df(t)	idf(t)	tf-idf
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network	3	500	$\log(10000/500) = 1.3$	3.9
the	50	9000	$\log(10000/9000) = 0.05$	2.5

Observations:

- "neural" appears less than "the" (5 vs. 50) but gets higher weight (10.0 vs. 2.5)
- Rare, discriminative terms weighted higher
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D2	$1 \times \log(3/2) = 0.18$	0	$1 \times \log(3/1) = 0.48$	0
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How to measure similarity between vectors?

Cosine Similarity Formula

$$\text{sim}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{|\vec{q}| \times |\vec{d}|} = \frac{\sum_t q_t \times d_t}{\sqrt{\sum_t q_t^2} \times \sqrt{\sum_t d_t^2}}$$

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Properties:

- Range: [-1, 1] (or [0, 1] for positive weights)
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- -1 = opposite vectors (not relevant for IR)

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Cosine Similarity: Example Calculation

Query: "cat dog" $\rightarrow \vec{q} = [0.5, 0.5, 0, 0]$
Documents:

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Similarity Calculations:

$$\begin{aligned}
\text{sim}(q, D1) &= \frac{(0.5 \times 0.35) + (0.5 \times 0.18)}{\sqrt{0.5^2 + 0.5^2} \times \sqrt{0.35^2 + 0.18^2}} \\
&= \frac{0.265}{0.707 \times 0.394} = 0.95 \\
\text{sim}(q, D2) &= \frac{0.5 \times 0.18}{0.707 \times 0.511} = 0.25 \\
\text{sim}(q, D3) &= \frac{0.5 \times 0.18}{0.707 \times 0.967} = 0.13
\end{aligned}$$

Ranking: D1 (0.95) ¿ D2 (0.25) ¿ D3 (0.13)

Vector Space Retrieval Algorithm

```
Input: Query q, Document collection D
Output: Ranked list of documents

Offline (Indexing):
for each document d ∈ D do
    for each term t in vocabulary do
        Compute w_{t,d} = tf(t, d) × idf(t)
    end for
    Store document vector d⃗
end for

Online (Query Processing):
for each term t in query q do
    Compute w_{t,q} = tf(t, q) × idf(t)
end for
Create query vector q⃗
for each document d do
    Compute sim(q, d) = cos(q⃗, d⃗)
end for
Sort documents by similarity (descending)
return Top-k documents
```

VSM vs. Boolean Retrieval

Aspect	Boolean	Vector Space
Matching	Exact (in/out)	Partial (weighted)
Ranking	No	Yes (by similarity)
Result size	Unpredictable	Top-k
Term weighting	Binary	TF-IDF
Query expression	Boolean operators	Term list
Robustness	Brittle	Graceful degradation
User friendliness	Expert users	General users
Relevance	All equal	Ranked by relevance

Modern Usage

- **Boolean:** Still used in legal, patent search (precision critical)
- **Vector Space:** Foundation for web search, most IR systems
- **Hybrid:** Many systems combine both approaches

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The Modern IR Stack

Today: Boolean + Vector Space (TF-IDF, Cosine)

↓

Lecture 2: Efficient Indexing (Sparse + Dense)

↓

Lecture 3: Better Ranking (BM25, LM, Query Expansion)

↓

Lecture 4: Neural Re-ranking (BERT, Transformers)

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- IR: Finding relevant unstructured documents at scale
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Due: Before Lecture 2

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Provided Materials

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- **Queries:** 50 test queries with relevance judgments
- **Starter code:** Python skeleton with TODOs
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Office Hours: [Your time]

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Course Website: [Your URL]

Start early on the assignment!

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Log-Frequency TF

$$\text{tf}(t, d) = \begin{cases} 1 + \log(\text{count}(t, d)) & \text{if count} > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Dampens effect of high term frequencies
- 10 occurrences not 10× more important than 1

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Identify and preserve named entities

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- "New York City" $\rightarrow$ single entity
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Compound Word Splitting

Handle compound words (especially in German, Dutch)

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- Language-specific rules

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- "Barack Obama" $\rightarrow$ single entity
- "New York City" $\rightarrow$ single entity
- Important for proper nouns

Compound Word Splitting

Handle compound words (especially in German, Dutch)

- "Informationssystem" $\rightarrow$ "Information" + "System"
- Language-specific rules

Backup: Advanced Preprocessing

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Case Folding Exceptions

Preserve case for certain terms

- Acronyms: "USA", "IBM"
- Names: "Bush" (president) vs. "bush" (plant)
- Context-dependent decision

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Backup: Alternative Similarity Measures

Euclidean Distance

$$\text{dist}(\vec{q}, \vec{d}) = \sqrt{\sum_t (q_t - d_t)^2}$$

- ✗ Sensitive to document length
- ✗ Not normalized
- Rarely used in IR

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Backup: Test Collections

Standard IR Test Collections

- **TREC:** Text REtrieval Conference collections (various domains)
- **Cranfield:** Classic small collection (1400 docs, aerodynamics)
- **CACM:** Computer science abstracts (3200 docs)
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2. **Topics/Queries:** Information needs
3. **Relevance judgments (qrels):** Which docs are relevant to which queries
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